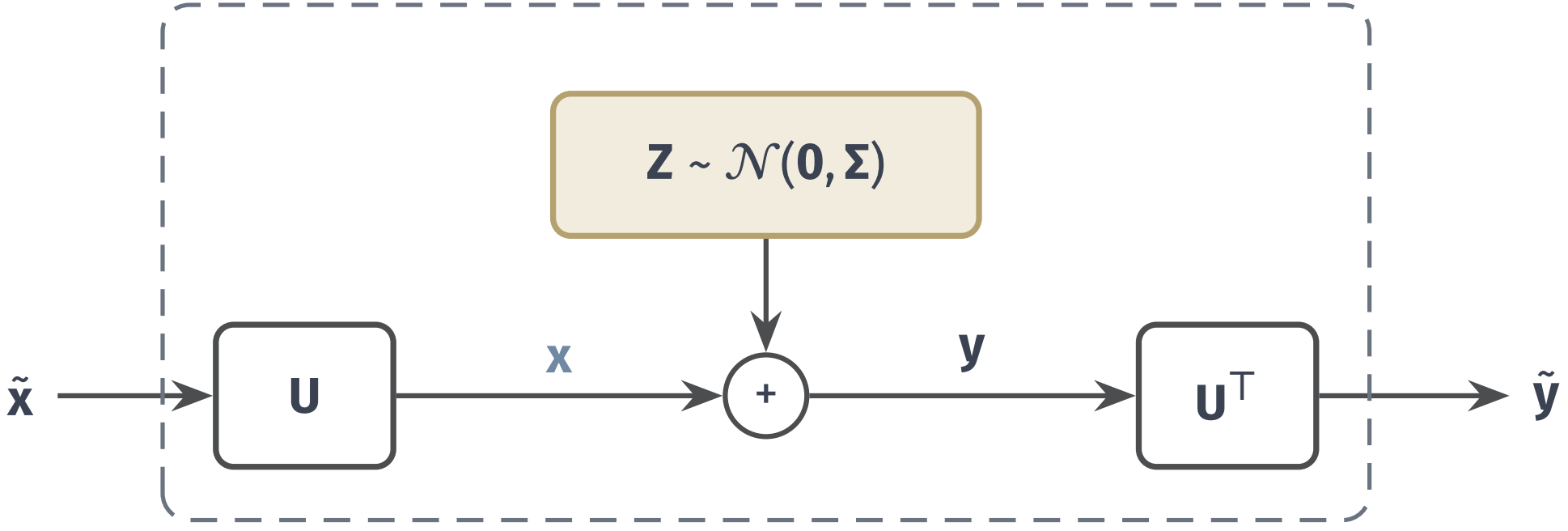

等价信道：$\tilde{\mathbf{y}} = \tilde{\mathbf{x}} + \tilde{\mathbf{Z}}$，$\tilde{\mathbf{Z}} = \mathbf{U}^{\mathsf{T}}\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Lambda})$

$$\|\mathbf{x}\|_2 = \|\tilde{\mathbf{x}}\|_2$$